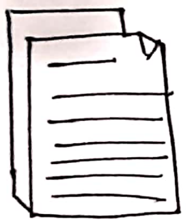
Vector databases — How it works.

Stores embedding (Vectors) and finds the most similar using nearest Neighbor search.

① Ingest Document



② chunk

↓

Chunk 1

Chunk 2

Chunk 3

⋮

Chunk N

③ Create Embedding

[0.21, -0.88, 0.44...]

[0.11, 0.35...]

⋮

[0.73, -0.12, 0.91]

④ Store in VD

↓

Vector Database
- store Vector
- Build index
- Enable fast simaliarity search

Using ANN (Approximate Nearest Neighbor)

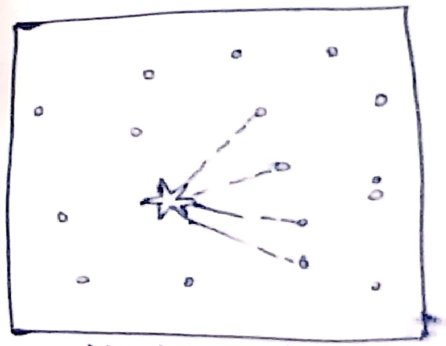
- HNSW
- IVF
- PQ/SQ
- etc

5 User Query → 6 Create Query Embedding → 7 Search → 8 Return Top K Most Similar

What is Retrieval Augmented Generation

[0.20, -0.85, 0.46.....]

↓
Nearest Neighbour



Vector space

1) chunk 7 (score 0.92)
2) chunk 3 (score 0.89)
3) chunk n (score 0.87)
⋮

↓

9 Send to LLM (with context)

Top K chunks
+
User chunks

← LLM Generates Answers

Why Vector DB?

- ✓ Semantic Search (meaning based)
- ✓ Handles unstructured data
- ✓ Scalable & fast with ANN
- ✓ Core Infra for Rag, semantic Search, Recommendations, Chatbots etc.

Similarity Measures

- Cosine Similarity (most common)
- Dot Product
- Euclidean Distance

Popular Vector Databases

1. Pinecone → Fully managed.
2. Weaviate → open source + hybrid search
3. Faiss → super fast